Smarter Device Financing Decisions

A better risk model and a simple in-store tool for Verizon

Proposal to the Verizon Steering Committee

November 2026 | The Outliers | Costantinos, Obaro, Stephen

Executive summary: Verizon is turning away customers it would profit from; our model and tool can finance them safely

+$21M

More profit per 1 million applications (+17%)

Today's denied and counteroffer applicants default at about 17%, far below Verizon's 41.7% break-even.

62%

A sharper risk model that stays explainable

Picks the riskier of two customers 62% of the time (today's score: 59%) and gets today's default rate right.

3

A simple in-store answer: approve, test or deny

Four facts reps already collect give one consistent decision in seconds. No more counteroffers.

Our ask: approve a 6-month pilot in 10% of stores, after a 2-month test behind the scenes.

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

What we heard: since May 2024, more customers default and Verizon has tightened, but the decision process has not changed

The situation

- **300,000 applications** in 2023 and 2024, decided in store
- Reps sort each applicant **by the internal risk score:** approve, counteroffer or deny
- May 2024: defaults rose **6.8%** → **10.7%** for the same kind of customer
- Verizon tightened: **approvals 89%** → **71%**
- **18% of applicants** now leave without a plan

What Verizon needs from this proposal

1. **A better risk model**
   More accurate than the current score, and right for today's conditions
2. **A live, low-risk decision tool**
   Fast and consistent in every store, and safe to roll out
3. **The best return on investment**
   A credible profit estimate for 1 million applications

The question: how many more customers can Verizon safely finance, and how should stores decide?

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

Each plan earns 25% of the device price if repaid and loses 35% if not, so a plan only loses money above 41.7% default risk

Four outcomes of every decision ($798 phone)

	Customer pays	Customer defaults
Offer plan	**+$200** Profit: 25% of price	**−$279** Loss: 35% of price
Deny plan	**−$200** Missed profit (opportunity cost)	**$0** Loss avoided

- **Offer whenever p < 0.25 ÷ 0.60 = 41.7%**
- **Counteroffers** lose 35–50% of customers and don't lower default

Expected profit per plan, $798 phone, today's conditions



Judge approvals by profit, not the default rate: approving more raises both defaults and profit.

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

We chose a simple scorecard: as accurate as a complex model, and easy to run and explain in a store

Today's internal score — 59%

- Doesn't adjust to today's higher defaults
- Rewards existing customers, who default more

Complex "black box" model — 62%

- Hard to explain to customers and regulators
- No more accurate than our scorecard

Our scorecard (chosen) — 62%

- Each factor has a clear effect that can be checked
- Gives each customer's chance of default, to compare with 41.7%

% = how often the model rates a customer who defaulted as riskier than one who paid

What goes in, and what the data showed

Credit score
Lower score, higher risk; little extra effect above 650

Income
U-shaped: risk is higher at both the lowest and highest incomes

Age
Two steps: 11.9% default under 25, 7.5% at 25–66, 9.4% at 67+

Existing customer
Default more (9.1% vs 7.2%), so no score bonus

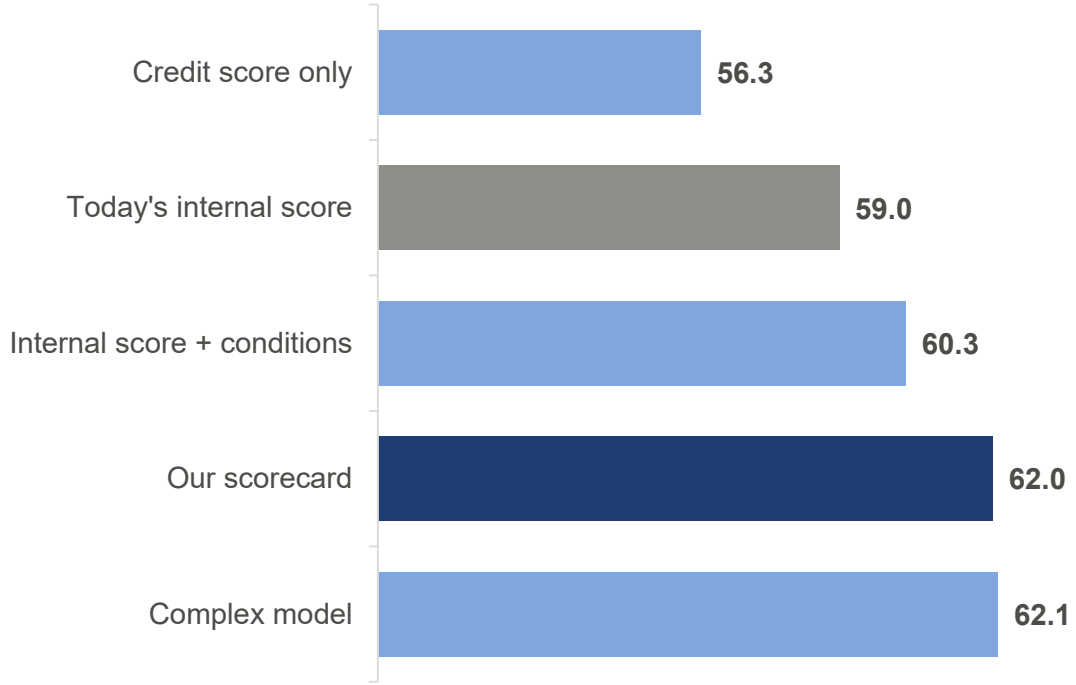
Current conditions
Defaults rose from 6.8% to 10.7% in May 2024; updated quarterly

Verizon gets top accuracy from a model every rep, manager and regulator can follow.

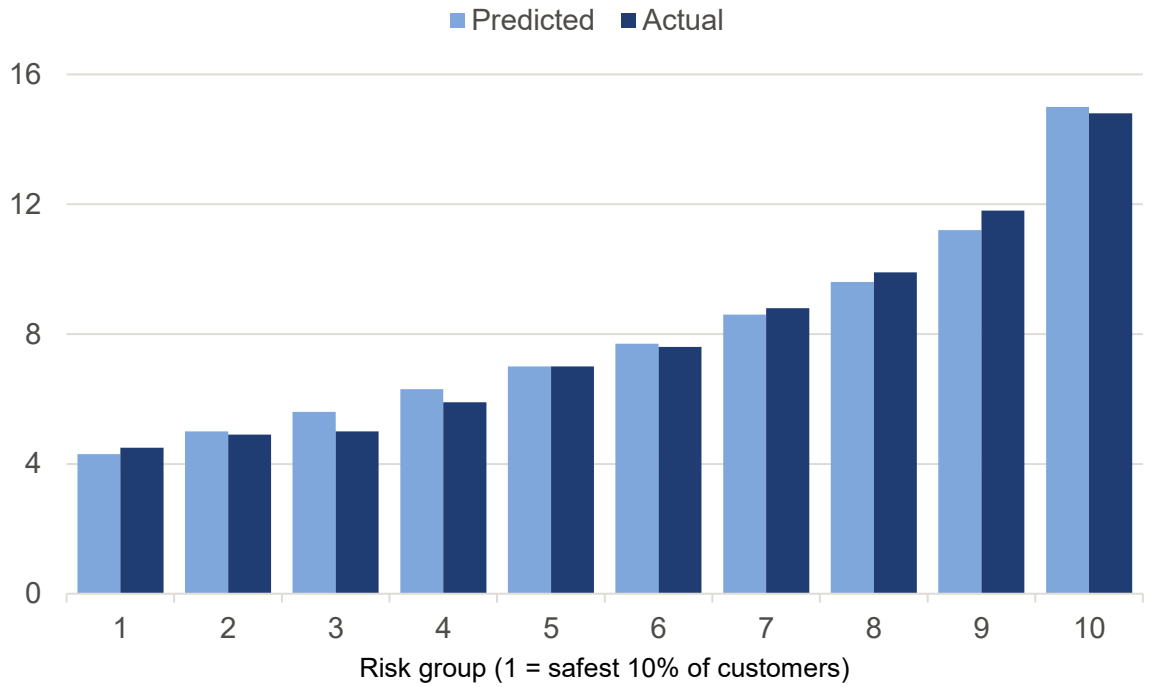
Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

Tested on 79,518 plans it had never seen, the scorecard sorts risk best and predicts today's default rate correctly

How often each model rates the defaulter as riskier (%)



Predicted vs actual default rate (%), in 10 equal-size risk groups



For today's conditions it predicts 10.7% default, matching actual; the current score predicts 8.6%.

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

One risk estimate replaces three score bands and the counteroffer step

Today (since May 2024)

71% — **Approve**
Requested phone and terms

17% — **Counteroffer**
Cheaper phone or bigger down payment; 40% walk away

11% — **Deny**
No outcome data, so no way to learn

Proposed: decide on each customer's chance of default

~97.6% — **Approve on requested terms**
Up to 25%: the range Verizon already finances

~2.4% — **Test lane**
25% to 41.7%: approve a random 10% to learn

<0.1% — **Deny**
Above the 41.7% break-even

Financed applicants rise from 82% to about 98%, and no one is pushed into a phone they didn't ask for.

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

In store, the rep enters four facts and gets one clear answer with the reasons behind it

Device financing check

Credit score	**575**
Annual income	**$42,000**
Age	**23**
Existing customer	**Yes**

APPROVE on the requested phone

Estimated default risk **18%** (approve up to 25%)

Main factors: under 25, credit score 575, existing customer

Why this is a low-risk system to run

1. **Uses data stores already collect**
   Credit score, income, age and customer status; no new checks
2. **Consistent and fast**
   One formula, the same answer in every store, in under a second
3. **Explainable**
   Every decision shows its main factors, for the rep, the customer and audits
4. **Controlled centrally**
   Cut-offs and the conditions term change at head office, with no store retraining

Same answer in every store, in seconds, with the reasons logged for every decision.

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

Today Verizon misses about $30M of profit per 1 million applications by turning away customers who would have paid

Today, per 1 million applications

	Would pay	Would default
Approved	**+$143.45M earned** 731.7k customers	**−$23.78M lost** 87.5k customers
Turned away	**$29.72M missed profit** 147.6k customers	**$9.33M loss avoided** 33.2k customers

Earned − lost = **$143.45M − $23.78M = $119.67M profit**

Our proposal, per 1 million applications

	Would pay	Would default
Approved	**+$172.28M earned** 863.3k customers	**−$31.86M lost** 114.1k customers
Turned away	**$3.22M missed profit** 16.0k customers	**$1.85M loss avoided** 6.6k customers

Earned − lost = **$172.28M − $31.86M = $140.42M profit**

How to read it: each box counts customers by what Verizon decided and what the customer did (or would have done). Missed profit is what Verizon would have earned from customers it turned away who would have paid.

Our proposal cuts missed profit from $29.7M to $3.2M, while losses on defaults rise by $8.1M.

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis; outcomes for turned-away customers are estimates (no repayment data exists)

Per 1 million applications, the new process adds about $21M of profit (+17%)

$$\Delta Value = \Delta Profit - \Delta Loss$$

$$= \Delta(Paying\ Customers) \cdot (Customer\ LTV) - \Delta(Defaulting\ Customers) \cdot (Cost\ of\ Default)$$

Per plan $= (1 - p) \cdot 0.25 \cdot Price - p \cdot 0.35 \cdot Price$

$(1 - p)$: chance the customer pays — $0.25 \cdot Price$: 25% of the price — p: chance the customer defaults — $0.35 \cdot Price$: 35% of the price

Simplified $= Price \cdot (0.25 - 0.60 \cdot p)$ break-even at p = 0.25 ÷ 0.60 = 41.7%

p = chance of default, from our scorecard *Price* = price of the phone the customer asked for *LTV* = lifetime value (profit over the plan)

Per 1M applications vs today	Δ Paying	ΔProfit	Δ Defaulting	ΔLoss	ΔValue	Evidence
A. Fund counteroffer walk-aways on the phone they asked for (68k)	+58.0k	+$11.8M	+10.0k	$2.8M	**+$8.9M**	Data now
B. Give accepted counteroffers the phone they asked for (105k)	same	+$2.3M	same	$0.6M	**+$1.7M**	Data now
C. Approve denied applicants up to 25% risk (90k)	+73.6k	+$14.7M	+16.6k	$4.6M	**+$10.1M**	Pilot confirms
Total	**+131.6k**	**+$28.8M**	**+26.6k**	**$8.1M**	**+$20.8M**	

Profit per 1M applications

$119.7M → $140.4M

+17%, about $21 per application

Default rate of financed plans

10.7% → 11.7%

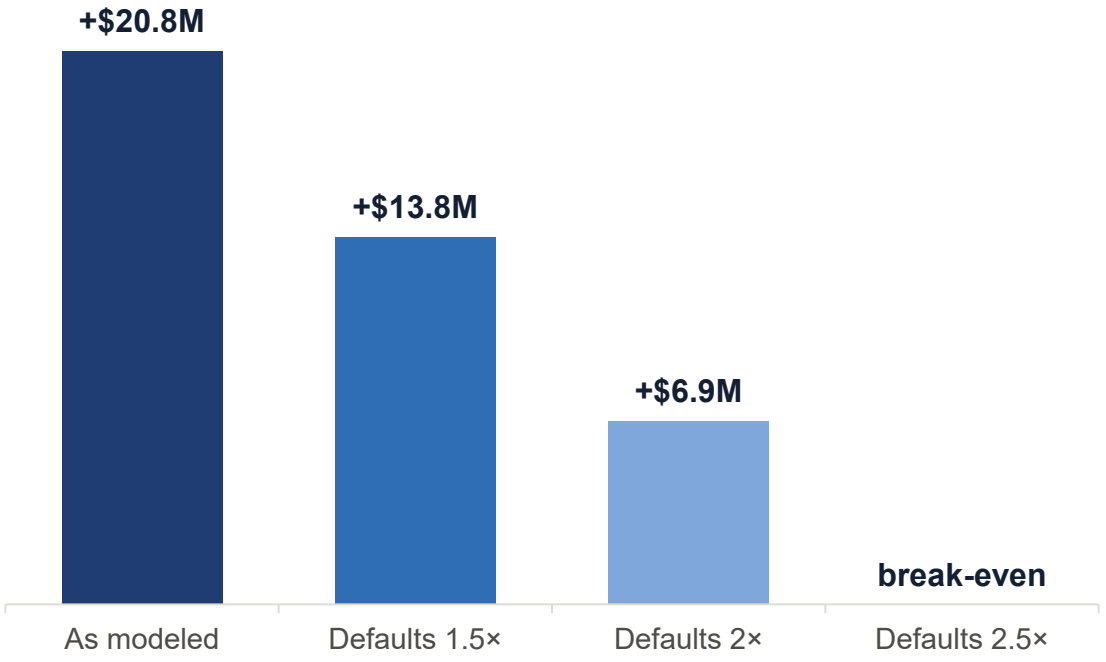
Not counted: service revenue on new lines

Losses rise $8.1M, but profit rises $28.8M. $10.7M is supported by the data today; the pilot confirms the rest.

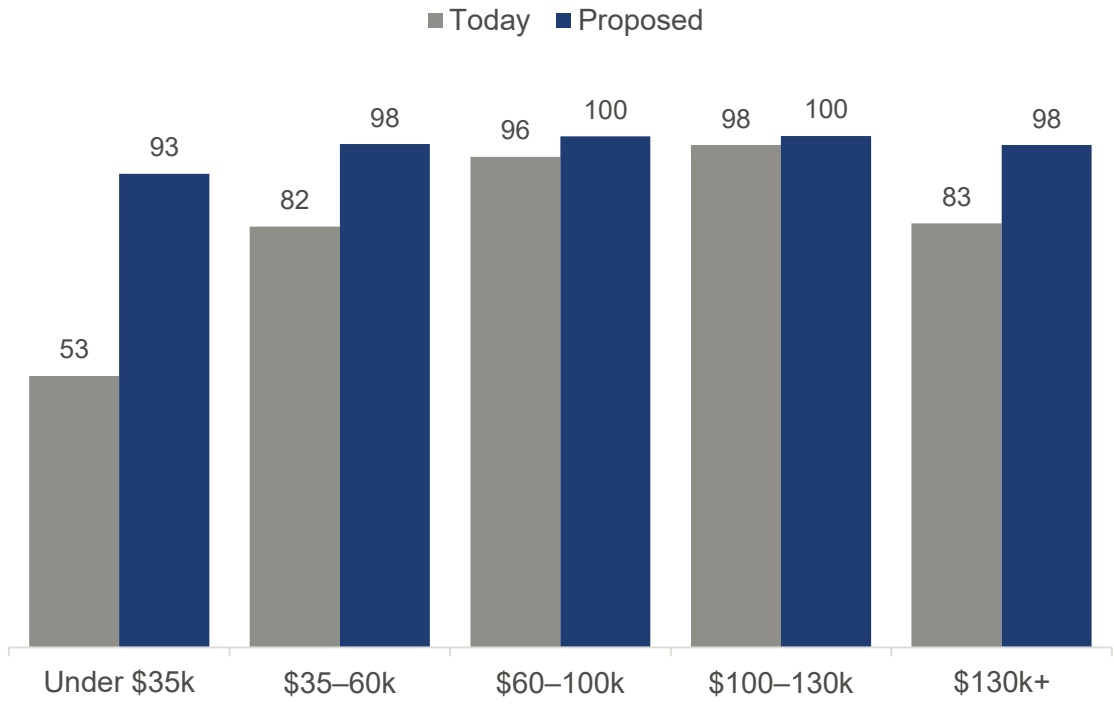
Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis; figures rounded to $0.1M, so rows may not add exactly

The gain holds even if defaults double, and lower-income applicants gain the most

Added profit per 1M applications, if defaults are higher



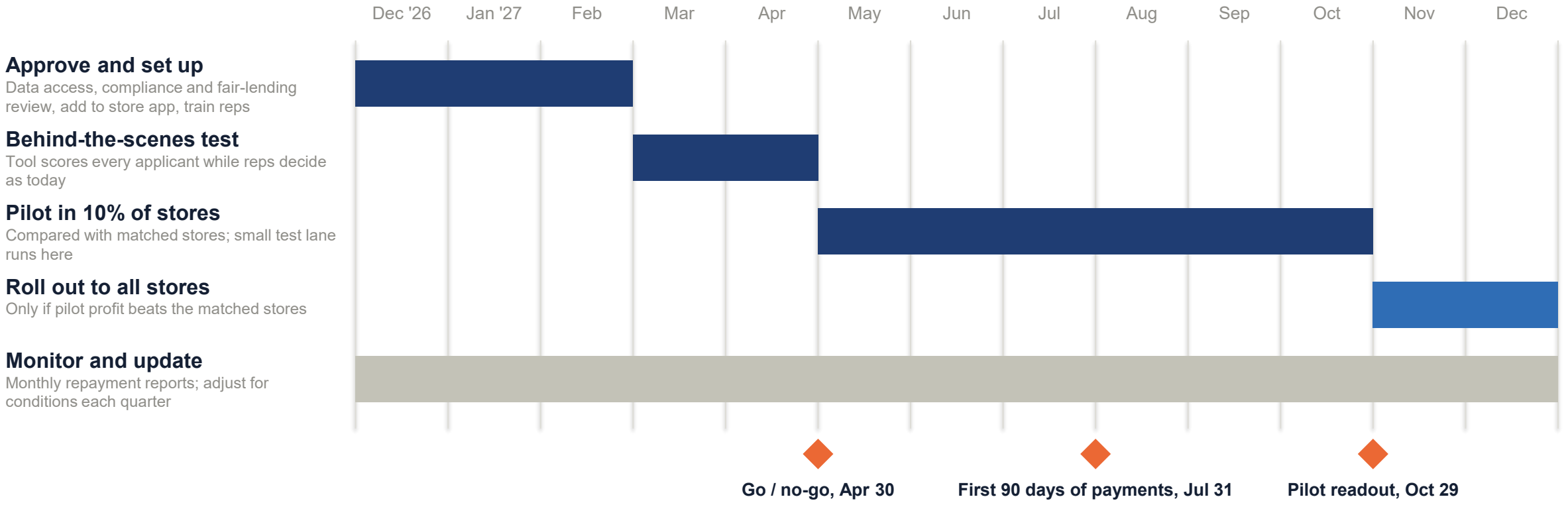
Share of applicants financed (%), by annual income



Profit only turns negative if new customers default 2.5× more than the model expects.

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

A 12-month plan proves the tool in 10% of stores before it reaches every store



Guardrail: pause new approvals if their missed payments run well above plan (expected default: about 17%).

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

Next steps: what we need from Verizon to start in December

+$21M

Clear value: per 1M applications, still positive if defaults double

12 months

Safe path: tested behind the scenes, then in 10% of stores

4 inputs

Explainable and fair: no demographic data; reasons shown for every decision

Action	Owner	By
Approve the pilot; name owners in Credit Risk and Retail Operations	Steering Committee	Dec 18, 2026
Share application volume and how profit per plan is calculated	Verizon Finance	Jan 15, 2027
Compliance and fair-lending review of the scorecard	Compliance, with our team	Feb 12, 2027
Add the scorecard to the store sales app; train pilot-store reps	IT and Retail Operations	Feb 26, 2027
Go / no-go after the behind-the-scenes test	Steering Committee	Apr 30, 2027
Pilot readout and decision to roll out	Steering Committee	Oct 29, 2027

Today's ask: approve the pilot and name an owner in Credit Risk and in Retail Operations by December 18.

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

Appendix

A. Model details B. Value calculation C. Fairness and definitions D. Assumptions

Appendix A: what drives risk in our scorecard, and what the data shows

Input	How it changes the chance of default	What the data shows
Credit score	Lower score, higher risk; little extra effect above 650	Today's score barely separates customers above 650
Income	Higher at the lowest and the highest incomes	Default 10.1% under $25k, 6.8% at $100–130k, 11.4% at $175k+
Age under 25	Higher	Default 11.9% at 18–24 vs 7.5% at 25–66; sharp step at 25
Age 67 and over	Higher	Default 9.4% at 67+; step at 67
Existing customer	Higher (today's score gives a bonus instead)	Existing customers default more: 9.1% vs 7.2%
Current conditions	Higher since May 2024; updated each quarter	Default rose from 6.8% to 10.7% for the same kind of customer
Tested and left out	Down payment, phone tier, today's internal score	Added nothing once the inputs above were included

Built on 185,542 financed plans from Jan 2023 to Dec 2024 and tested on 79,518 plans it had not seen. It picks the riskier of a defaulting and a paying customer 62% of the time (today's score 59%, credit score alone 56%); within May–Dec 2024 alone, 57.8% vs 57.2%.

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

Appendix B: how the $21M per 1 million applications is calculated

Item	May–Dec 2024	Per 1M applications	If defaults 2×
Applications	100,280	1,000,000	
Profit today (actual)	$12.0M	$119.7M	
Counteroffer group on requested terms (6,815 new + upgrades)	+$1.07M	+$10.7M	+$4.8M
Denied applicants up to 25% chance of default (9,048)	+$1.01M	+$10.1M	+$2.1M
Total added profit	+$2.08M	+$20.8M (+17%)	+$6.9M
Share of applicants financed	81.9% → 97.7%		
Default rate of financed plans	10.7% → 11.7%		
Average chance of default, newly approved customers	16.8%		

Per 1M = May–Dec 2024 × 9.97 (1,000,000 ÷ 100,280). Each added plan is valued at Price × (0.25 − 0.60 × chance of default). Assumptions are listed in Appendix D.

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis; figures rounded to $0.1M, so rows may not add exactly

Appendix C: fairness check and definitions

Group (self-reported)	Financed today	Proposed	Predicted vs actual default
Black	73.9%	95.8%	8.5% vs 8.8%
Hispanic	78.8%	96.9%	8.2% vs 7.8%
White	84.3%	98.3%	7.9% vs 7.8%
Female	81.9%	97.8%	8.0% vs 7.9%
Male	81.9%	97.7%	8.0% vs 8.1%

Predicted vs actual: on plans the model had not seen. Race, ethnicity and gender are not model inputs; they are used only for this check.

Term	Definition
Financed	Approved, or accepted a counteroffer (265,060 plans)
Not financed	Denied, or declined the counteroffer (34,940)
Periods	Baseline Jan–Jul 2023; promotion Aug 2023–Apr 2024; current May–Dec 2024
Profit rule	+25% of price if repaid, −35% if defaulted (every row)
Ranking accuracy	How often a model rates a customer who defaulted as riskier than one who paid

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

Appendix D: the assumptions behind the status quo and our estimate

Assumption	What we used	Why it is reasonable
Status quo	Verizon's actual decisions and results, May–Dec 2024	Most recent period, after the May 2024 tightening
Profit per plan	+25% of price if repaid; −35% if the customer defaults	Matches the profit column in every row of the data
Turned-away customers	Chance of default from our scorecard: 18% on average	No repayment data exists; tested at 1.5× and 2×, loses money only at 2.5×
Acceptance	Customers approved on requested terms take the plan	Every approved applicant in the data was financed
Price	Average price of the phone tier requested	Prices are fixed per tier in the data
Scale	Results grow in proportion to 1 million applications	Same mix of applicants as May–Dec 2024
Not counted	Service revenue on new lines, tool costs, test-lane results	Each $1M of tool cost is repaid by about 48,000 applications

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis